OpenAPI Documentation

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OpenAPI Documentation

Scope

How the API contract is described, published and kept honest.

Official requirements

Subject V.4 requires reference documentation presenting the API's methods, inputs and outputs, and names Swagger as an acceptable generator. Peer evaluators and the team's own mobile developers are the first consumers, so the document is graded material.

Project policy

- Generator: `Microsoft.AspNetCore.OpenApi`, the built-in one (`brain/decisions/ADR-0021-backend-dependency-set`). Not Swashbuckle's `SwaggerGen` — the reference repository uses it because the built-in generator did not exist in net6. **One generator describes the API.** Two would mean two documents for one surface, and the committed `docs/api/openapi.json` would stop being the contract the moment they disagreed.
- The OpenAPI 3.1 document is **generated from the code**, never hand-maintained. A hand-written spec drifts and drift is worse than no spec.
- Interactive UI is served in Development, by `Swashbuckle.AspNetCore.SwaggerUI` pointed at `/openapi/v1.json` — the UI middleware alone, never the metapackage that also brings `SwaggerGen` (`brain/decisions/ADR-0035-swagger-ui-for-the-openapi-document.md`). In other environments the raw document is served and the UI is not registered: an unauthenticated catalogue of every endpoint is a map, and `API8` in the API checklist is about exactly this.
- **The UI is a reader, not a source.** A wrong endpoint on the page is a wrong endpoint in the document, which is a wrong attribute in the code. The fix is never in the UI configuration.
- Every endpoint documents: summary, description, all parameters, request body schema, **every** response status it can return, and at least one realistic example.
- The Problem Details error schema is declared once and referenced by every error response.
- Security schemes are declared (bearer JWT) and applied per operation. Endpoints that are intentionally anonymous say so explicitly.
- Enums appear as enums in the schema, with their string or integer values, not as bare integers.

- Pagination envelope and client context headers are declared as reusable components.
- The generated document is exported to `docs/api/openapi.json` on release and committed, so a peer evaluator can read the contract without running the service.
- CI fails if the generated document does not build, and a diff in `openapi.json` must appear in the pull request that changed the endpoint.

XML comments

Enable `GenerateDocumentationFile`; use `<summary>`, `<param>`, `<response>` on controller actions so the description comes from the source rather than a parallel document.

Checklist

- ☐ Document generates and validates in CI
- ☐ Every endpoint has all response codes and one example
- ☐ Security scheme declared and applied
- ☐ `docs/api/openapi.json` updated in the same PR as the endpoint change
- ☐ Interactive UI registered only inside the Development branch

References

- OpenAPI 3.1 specification — https://spec.openapis.org/oas/v3.1.0
- ASP.NET Core OpenAPI support — https://learn.microsoft.com/aspnet/core/fundamentals/openapi/overview

Related

`../rest-api-design/STANDARD.md` · `../../70-delivery/ci-cd-containerization/STANDARD.md`